COMPUTER MEMORY

RAM, ROM, PROM, EEPROM and how to check your memory type – explained step by step



What we will learn

1 Memory basics

Main vs secondary memory, volatile vs non-volatile

2 RAM

What it is, how it works and its types (DRAM, SRAM, DDR)

3 Check RAM type

wmic commands and the DDR code numbers

4 ROM

Read-only memory, firmware and BIOS

5 PROM

Programmable ROM – written once by the user

6 EEPROM

Electrically erasable ROM – easy to modify

Main memory vs secondary memory

MAIN (PRIMARY) MEMORY

RAM, ROM and cache. Directly used by the CPU. Fast, small and expensive per GB. RAM is volatile; ROM is non-volatile.

Analogy: the desk you work on

SECONDARY MEMORY

HDD, SSD, pen drive, CD/DVD. Permanent storage, large and cheap per GB, but slower. Data is first loaded into RAM.

Analogy: the cupboard where files are kept

Volatile vs non-volatile

1 Volatile

Data is lost when power is switched off. Example: RAM and cache. Like writing on a whiteboard.

2 Non-volatile

Data stays after power off. Example: ROM, HDD, SSD, pen drive. Like writing in a notebook.

3 Why it matters

RAM needs power to hold data, so unsaved work is lost in a power cut. Save files to secondary memory.

4 Where startup code lives

ROM is non-volatile, so the startup instructions (BIOS) are always there when you switch on.

RAM – Random Access Memory

1 What it is

The main memory of the PC. It holds the programs and data the CPU is using right now.

2 Random access

Any location can be reached directly in the same short time, unlike a tape that is read in order.

3 Volatile

Contents disappear when power is off, so work must be saved to the HDD/SSD.

4 Where it sits

In the memory (RAM) slots on the motherboard – usually 2 to 4 slots; more on high-end boards.

RAM – Types and capacity

1 DRAM

Dynamic RAM: needs constant refreshing; cheap and dense. Used as the main RAM sticks.

2 SRAM

Static RAM: faster, no refresh, but costly. Used as cache inside the CPU.

3 SDRAM / DDR

Synchronous RAM that works with the system clock. DDR (1, 2, 3, 4) doubles data per clock tick.

4 DIMM / SODIMM

DIMM sticks are for desktops; the smaller SODIMM is for laptops.

RAM – Working steps

1 Load

When you open a program it is copied from the HDD/SSD into RAM.

2 Work

The CPU reads and writes RAM directly while you work. RAM is much faster than a disk.

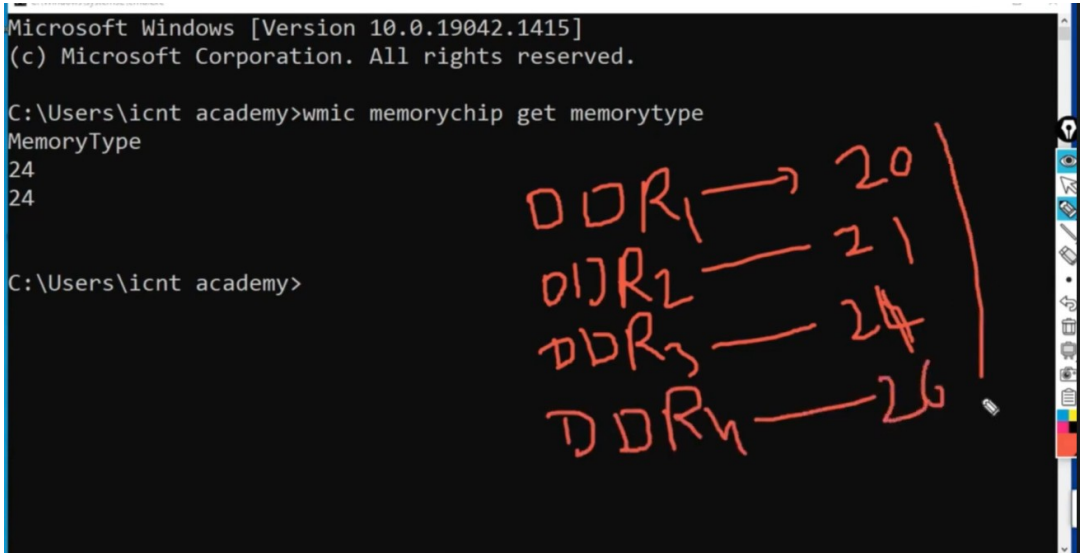
3 Save

Saving copies data from RAM back to secondary memory.

4 Clear

On shutdown or power cut, RAM is wiped clean.

Check RAM type with a command



DDR code in MemoryType

- DDR1 → 20
- DDR2 → 21
- DDR3 → 24
- DDR4 → 26

wmic memorychip get memorytype

```
C:\Users\icnt academy>wmic memorychip get memorytype
MemoryType
24
24
```

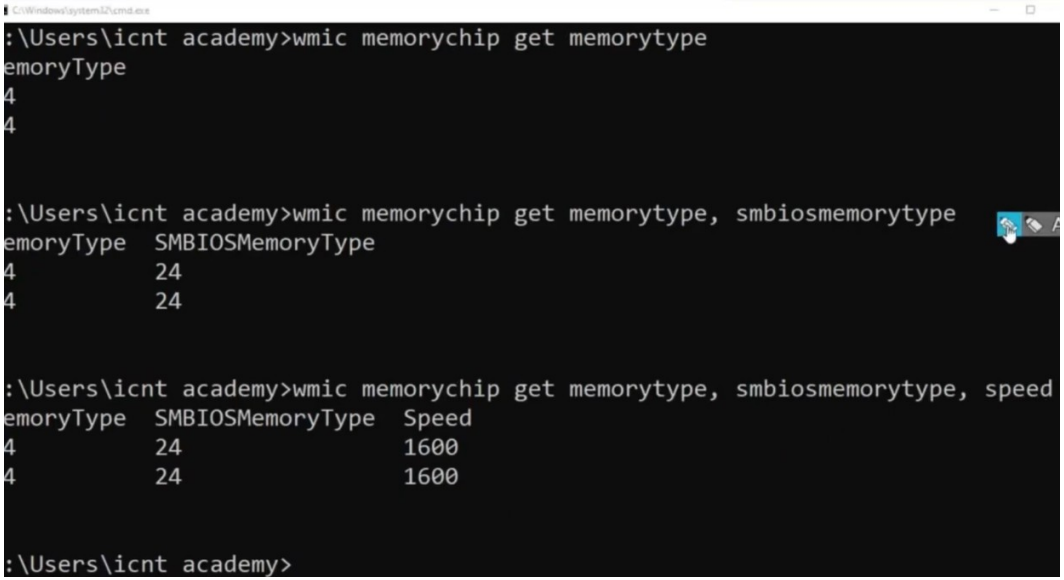
1 Two lines = two sticks

Each output line is one RAM stick. Two lines of 24 means two memory modules are installed.

2 24 means DDR3

From the chart: 20 = DDR1, 21 = DDR2, 24 = DDR3, 26 = DDR4. So both sticks are DDR3.

More memory commands



MemoryType
Older code for the type of memory.

SMBIOSMemoryType
Type read from the BIOS: 24 = DDR3.

Speed
1600 means 1600 MHz, so these are DDR3-1600 sticks.

```
wmic memorychip get memorytype, smbiosmemorytype
wmic memorychip get memorytype, smbiosmemorytype, speed
```

ROM

FROM THE LECTURE

- Read only memory
- Also known as firmware
- Data stored in this memory cannot be modified.
- Non-volatile memory
- ROM usually stores the startup instructions
- When computer is switched on , the first piece of instruction of a program accessed by the CPU is BIOS which is a ROM pre-programmed to start the computer operations

ROM – Explained

1 Read only

The CPU can read it, but normal use cannot change it.

2 Firmware

Permanent software stored inside a chip, so it is called firmware.

3 Non-volatile

Keeps data without power – so startup code is always ready.

4 BIOS starts the PC

On power-on the CPU first runs the BIOS, a pre-programmed ROM that starts the computer.

PROM

FROM THE LECTURE

- Programmable read-only memory (PROM), or one-time programmable ROM (OTP),
- Empty of data when the chip is manufactured,
- can be programmed by the user.
- Once programmed the data cannot be erased.



NEC PROM chip

PROM – Explained

1 **Programmable ROM**

Also called OTP – one-time programmable ROM.

2 **Blank at first**

The chip leaves the factory empty.

3 **User programs it**

The user writes the data using a special device (a programmer).

4 **Permanent**

After writing, the data cannot be erased or changed.

EEPROM

FROM THE LECTURE

- Electrically Erasable programmable read-only memory (EEPROM)
- Can be written to or programmed.
- can be erased with the help of electrical devices
- So data can be easily modified

EEPROM – Explained

1 Electrically erasable

Data is erased with electrical signals, without removing the chip.

2 Can be programmed

It can be written again and again.

3 Easily modified

Highlighted in the lecture: the main advantage over PROM.

4 Real use

Stores BIOS settings and firmware that must be updated later.

ROM, PROM and EEPROM compared

	ROM	PROM	EEPROM
Full name	Read only memory	Programmable ROM	Electrically Erasable PROM
Written by	Manufacturer	User, once	User, many times
Erasable?	No	No	Yes, electrically
Volatile?	Non-volatile	Non-volatile	Non-volatile
Use	Startup / BIOS	Custom permanent code	Updatable firmware

RAM vs ROM

	RAM	ROM
Full name	Random Access Memory	Read Only Memory
Volatile?	Yes – lost on power off	No – kept
Read / write	Both, fast	Mostly read only
Stores	Running programs and data	Startup instructions (BIOS)
Location	RAM slots on the board	Chip on the board

Thank you

RAM = fast, volatile working memory. ROM, PROM and EEPROM = non-volatile memory that keeps startup and firmware data.